Sana Qureshi

Data Scientist · Toronto, Canada · sana@example.com

I build models that decide things with money on the line: which payments look like fraud, how much stock a store should order, and whether a product change actually helped.

EXPERIENCE

Data Scientist, Fraud and Experimentation, Fernway Financial — Toronto · Mar 2022 to Present

- Own the real-time card fraud model that scores about 1.8 million payments a day in under 40 ms, and cut fraud losses by 31% in its first year without raising the decline rate.
- Built the statistics engine behind Fernway's A/B testing platform: CUPED variance reduction, sequential testing and automatic sample ratio checks, now used by 14 product teams.
- Run the monthly model review with risk and compliance, and wrote the playbook for retraining, drift alerts and rollbacks.

Data Scientist, Demand Forecasting, Harbour & Pine Grocers — Toronto · Jun 2019 to Feb 2022

- Replaced a spreadsheet forecast with a LightGBM model for 140 stores and 22,000 products, lowering weekly forecast error from 14.2% to 8.9% WMAPE.
- Worked with store managers on fresh produce ordering, which cut spoilage by 17% across the pilot region.

Research Assistant, Applied Statistics Lab, Lakemont University — Toronto · Sep 2016 to May 2019

- Studied forecast reconciliation for hierarchical retail data, which became my master's thesis and a journal paper.
- Taught the lab sections of an undergraduate regression course for four terms.

FLAGSHIP: LARKSPUR

Every card payment gets a fraud score in under 40 milliseconds, and every analyst decision makes the next score better.
Ingest → Features → Score → Review, with a shared label feedback loop. 1.8 million payments scored a day, p99 latency 38 ms. Fraud losses down 31% in the first year at the same decline rate. Shadow mode and a one-click rollback for every model release.

SELECTED PROJECTS

Account Takeover Detection 2025
Flags logins that look like someone else is using a customer's account. (74% recall at 1% false positives)

Merchant Risk Graph 2024
Finds risky merchants by the cards, devices and bank accounts they share. (38% more bad merchants caught)

CUPED Variance Reduction 2023
Makes A/B tests reach a decision faster by using each user's pre-experiment behaviour. (42% less variance per test)

Sequential Testing Guardrails 2024
Lets teams check results every day without fooling themselves. (5% false positives with daily peeking)

Grocery Demand Forecast 2021
Weekly sales forecasts for 140 stores and 22,000 products. (8.9% WMAPE, down from 14.2%)

Fresh Produce Ordering 2021
Turns probabilistic forecasts into daily order quantities for perishables. (17% less spoilage)

Small Business Cash Flow Forecast 2025
A 30-day balance forecast shown to Fernway's small business customers. (22% fewer overdrafts)

PAPERS AND TALKS

- Two-Hop Features for Merchant Fraud Detection at Small Scale. AMLF 2024, 2024. Best Paper Award.
- Faster Experiments with Fewer Users: CUPED in Payments. Northern Data Summit 2023, 2023.
- Reconciling Hierarchical Forecasts for Perishable Retail Demand. Journal of Retail Analytics, 2019.

EDUCATION

MSc in Statistics, Lakemont University, Toronto, 2016 to 2018.

SKILLS

Modelling: Gradient boosting, Time series, Survival analysis, Graph features, Uplift modelling, PyTorch, scikit-learn, LightGBM
Experimentation: A/B testing, CUPED, Sequential testing, Causal inference, Power analysis, Bayesian methods
Data and MLOps: SQL, PySpark, Airflow, dbt, Kafka, MLflow, Docker, Feature stores
Languages and tools: Python, R, pandas, statsmodels, Git, Jupyter, PostgreSQL

CERTIFICATIONS

Causal Inference for Practitioners, Lakemont University Continuing Studies (Oct 2024); Certified Experimentation Practitioner, Northern Analytics Guild (Apr 2023); Probabilistic Forecasting, Open Forecasting Academy (Nov 2020)

Sample portfolio. The person, companies and projects are fictional.